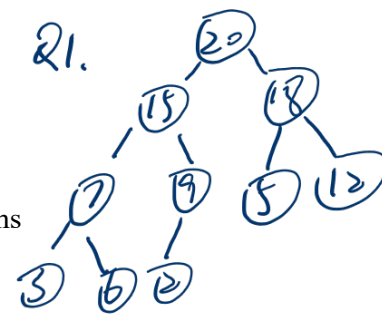


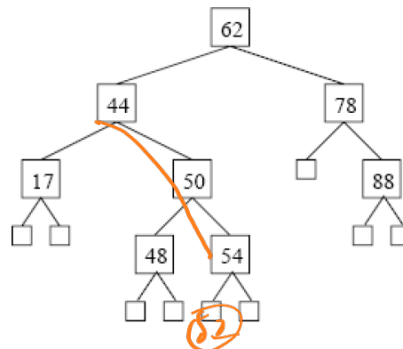
INT202 Complexity of Algorithms

Tutorial 3



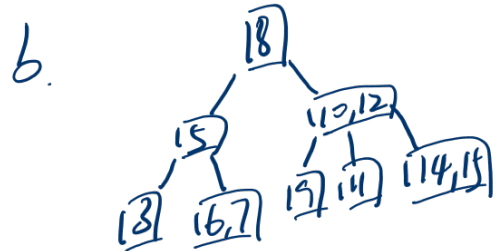
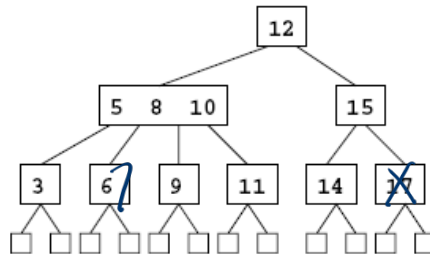
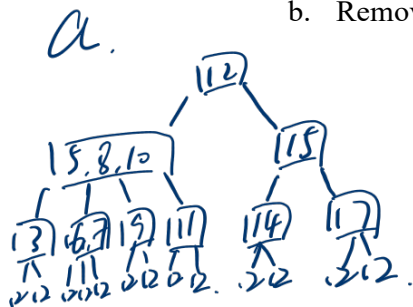
Q1. Show that array [20 15 18 7 9 5 12 3 6 2] forms a max-heap.

Q2. Draw the AVL tree resulting from the insertion of an item with a key 52 into the given binary tree. (Only the final tree is needed.)

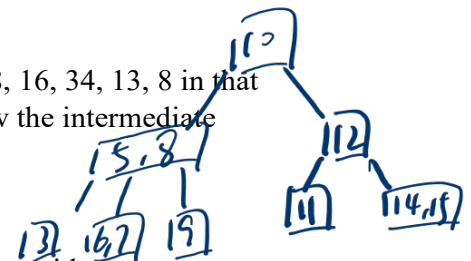


Q3. Let T be a (2, 4) tree shown below, which stores items with integer keys. Draw all the steps and the resulting trees obtained by performing the following operations on the original (given) T.

- Insert an item with key 7 into T.
- Remove an item with the key 17 from T.



Q4. Draw the binary min-heap that results from inserting: 77, 22, 9, 68, 16, 34, 13, 8 in that order into an initially empty binary min heap. You are required to draw the intermediate heaps after each insertion and the final heap.

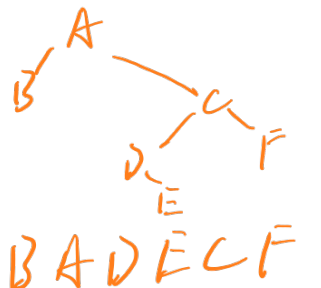


Q5. Insert the following sequence of elements into an AVL tree, starting with an empty tree: 10, 20, 15, 25, 30, 16, 18, 19. Draw the AVL tree after each insertion.

Q6. Given the following pre-order and in-order binary tree traversals, draw the binary tree.

Pre-order A, B, C, D, E, F

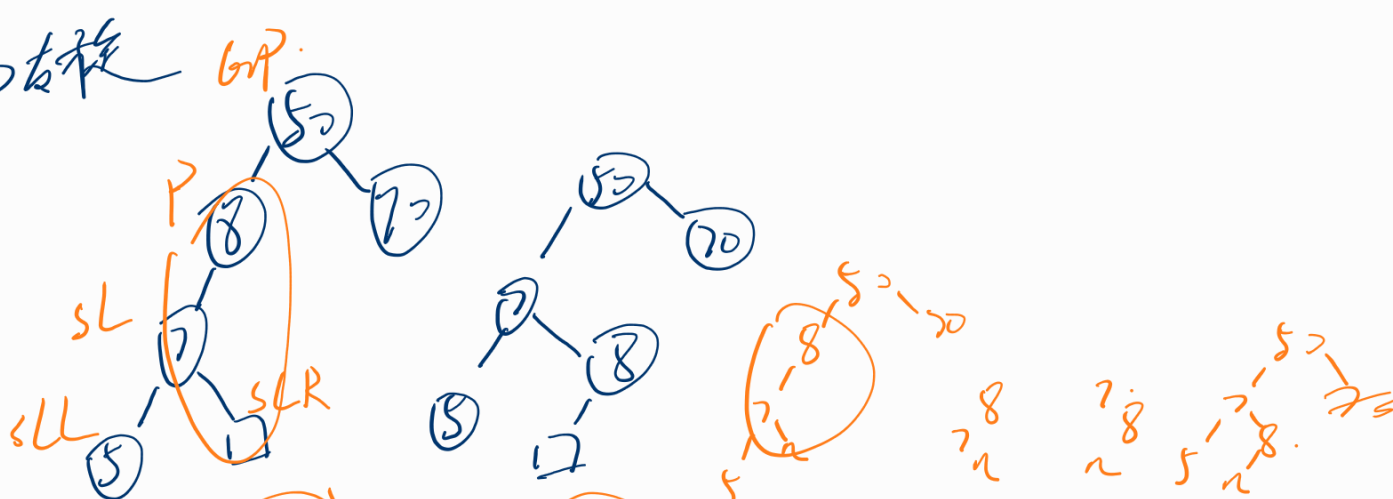
In-order B, A, D, E, C, F



右高 → 左旋



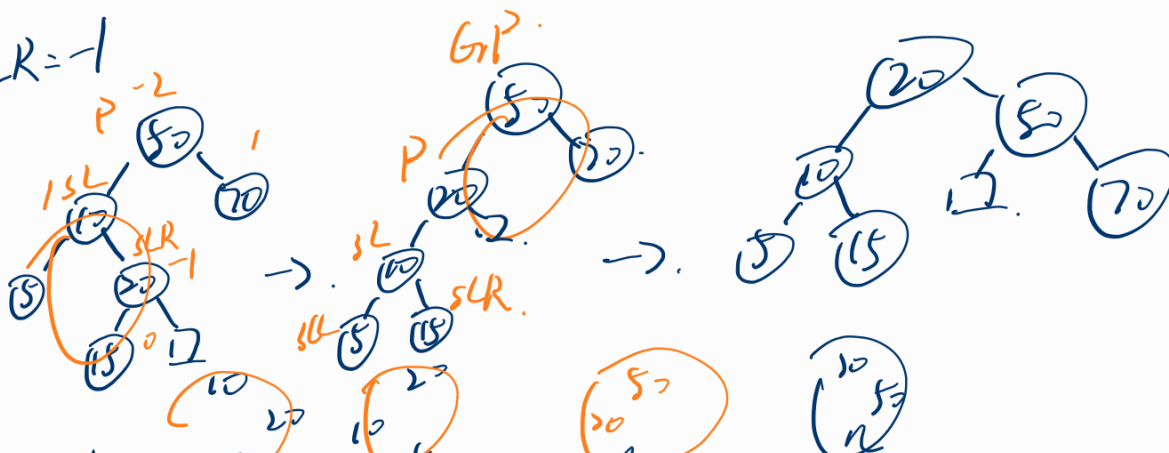
右高 → 左旋



左高 ① SLR = 0



② SLR = -1



③ SLR = 1

